INTEL CORPORATION

SAP Joule

Architecture Overview

Version: 1.1

Date: January 2026

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1. Executive Summary

Purpose

This document defines the **SAP Joule AI Architecture** for Intel's enterprise SAP landscape. SAP Joule is an AI-powered assistant that transforms how users interact with SAP applications through natural language, delivering transactional, navigational, informational, and analytical capabilities across Intel's SAP portfolio.

Scope

The architecture encompasses three deployment patterns:

- **Joule Assistant Core** – Central conversational interface for information, navigation, and analytics (powered by SAP AI Core and SAC Just Ask)
- **Embedded Agents** – SAP-delivered agents activated within LoB applications (Ariba, Concur, IBP, Fieldglass, Commerce Cloud, Signavio, APM, PAPM, BN4L)
- **Custom Agents** – Developer-built agents via Joule Studio for S/4HANA on-prem and cross-system orchestration (Skills mandatory for Studio orchestration, optional when using MCP/N8N)

Key Architectural Components:

- **SAP Build Work Zone** – Primary entry point / launchpad for Joule; navigation service, CDM content sync via IPS, and role-based access
- **SAP AI Core** – Generative AI Hub providing Dialog Management LLM and partner LLM ecosystem for Joule processing
- **SAP AI Launchpad** – AI management and monitoring interface
- **SAP Cloud Identity Services (IAS/IPS)** – Identity Authentication (SSO via SAML2/OIDC) and Identity Provisioning (SCIM-based user/group/role sync); common instance recommended across prod and non-prod
- **Scenario Catalog** – Metadata registry of all available Joule scenarios, functions, and skills across SAP cloud applications
- **Knowledge Catalog** – SAP-owned and customer-owned knowledge base driving Retrieval Augmented Generation for enterprise (RAGe)

- **Document Grounding** – RAGe-powered grounding service for informational queries against enterprise content
- **Response Filtering / Responsible AI** – Enterprise security, data privacy, and responsible AI guardrails applied before response delivery
- **SAP Build Process Automation** – SAP-native workflow and process automation (complements n8n for Intel-specific flows)

Note: S/4HANA Cloud (J4C) is a SAP-managed offering where custom agents cannot be built directly; SAP-delivered agents are used instead.

> **Intel-Specific Terminology:** IF = Intel Foundry | IP = Intel Corp / Products | CFIN = Central Finance

> **Other SAP Cloud Products:** SAP SuccessFactors, SAP Datasphere, SAP LeanIX, and SAP DSC are also Joule-enabled SAP cloud solutions depicted in the SAP reference architecture. These are not in current Intel scope but should be considered for future expansion as Joule capabilities mature across the SAP portfolio.

Key Architectural Principles

Principle	Description
Contextual Intelligence	Role-based, context-aware AI assistance within SAP applications
Native SAP Integration	Seamless access to SAP backends via Skills, Scenario Catalog, and Knowledge Catalog
Complementary Runtime	Joule handles SAP-native AI; ECA/Azure handles analytics; iGPT handles general-purpose AI
Governed Execution	Response filtering, responsible AI guardrails, and human-in-the-loop approval gates
Identity-First Security	SAP Cloud Identity Services (IAS/IPS) as unified identity foundation
Extensible Agent Model	Custom agents via Joule Studio with Skills, MCP, and A2A protocol support

Business Outcomes

- **Accelerated SAP User Productivity**: Natural language interaction reduces navigation and task completion time
- **Governed AI across SAP Portfolio**: Consistent security, audit, and responsible AI compliance across all LoB applications

- **Extensible Agent Ecosystem**: Custom agents via Joule Studio enable Intel-specific SAP process automation
- **Enterprise Knowledge Access**: RAGe-powered document grounding provides instant access to SAP and enterprise knowledge

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2. Key Capability Patterns

Per the SAP reference architecture, Joule's capabilities are categorised into four interaction patterns that define how users engage with the copilot:

Pattern	Description
Transactional	Direct entry point to SAP backend systems. Triggers and influences business processes via natural language and generative AI – e.g., approving purchase orders, creating job positions, or any CRUD-based interaction. All SAP Cloud products are developing content packages for this pattern.
Navigational	Helps users navigate directly to the relevant SAP application screen. Particularly valuable for users unfamiliar with SAP application navigation.
Informational	Provides knowledge-based results (e.g., policy questions). Grounded on SAP-owned content (SAP Help Pages) or customer-owned content via the Document Grounding service and RAGe (Retrieval Augmented Generation for enterprise).
Analytical	Delivers insights and data analysis. Users ask questions about business metrics, trends, and performance indicators; Joule returns relevant insights from SAP Analytics Cloud (Just Ask).

3. Characteristics (SAP Design Principles)

Characteristic	Description
Revolutionize User Experience	Contextual assistance, routine task automation, and insight delivery driving better business outcomes.
End-User Interactions	Single AI copilot accelerating every process across all SAP solutions.
Seamless Integration & Reliable Insights	Native access to enterprise data with role-based access control ensuring grounded, reliable recommendations.
Customizable & Extensible	Custom skills and agents via Joule Studio and SAP BTP, tailored to specific business needs.
Security & Compliance	Stringent security standards protecting sensitive business data throughout the interaction lifecycle.
Human-Centric Governance	Built-in responsible AI safeguards aligned with SAP AI ethics policy and UNESCO's 10 guiding principles on Ethics of AI.

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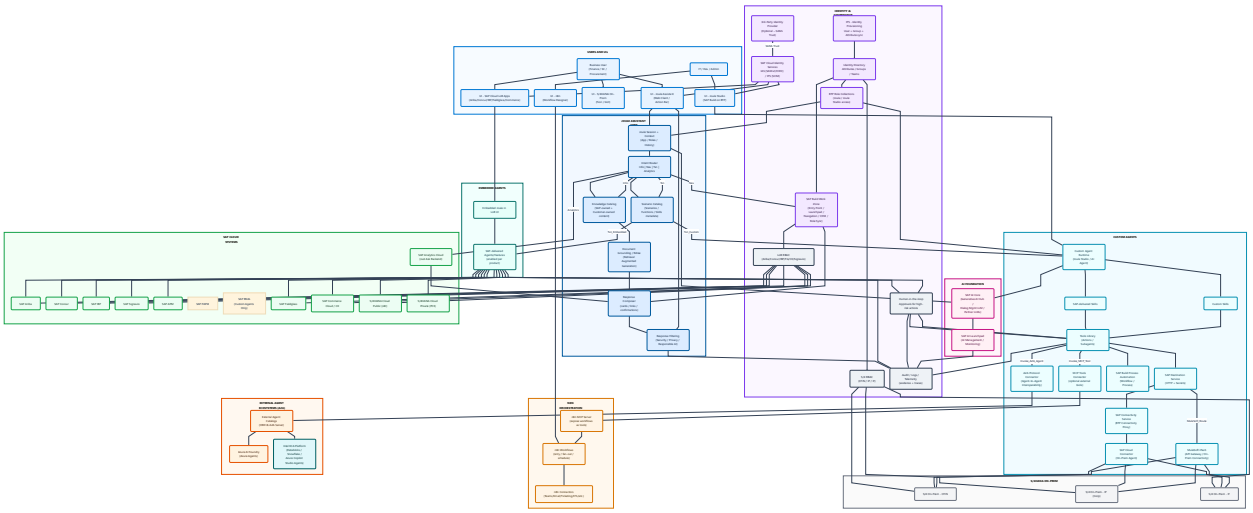
4. Architecture Overview

Architecture Overview – Joule Component Architecture Diagram



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5. Implementation Phases

Phase 1 – Foundation & Identity

Purpose: Establish the identity foundation, AI infrastructure, and prerequisites for Joule enablement across the SAP landscape.

Key Activities:

- Configure **SAP Cloud Identity Services** (IAS + IPS + Identity Directory) as unified identity foundation; use a common instance (prod and non-prod) with a single domain URL (`cloud.sap.com`) per SAP recommendation
- Configure IAS for global SSO using **SAML2/OIDC** trust relationships
- Set up IPS for **SCIM-based** user, group, and attribute synchronization
- Establish Identity Directory with attributes, groups, and teams
- Optionally configure **3rd-party Identity Provider** trust via SAML with SAP Cloud Identity Services
- **Configure SAP Build Work Zone** as Joule entry point / launchpad, navigation service, and CDM content sync
- **Set up SAP AI Core** (Generative AI Hub / Dialog Management LLM) for LLM foundation
- **Deploy SAP AI Launchpad** for AI management and monitoring
- **Define BTP subaccount topology**: Joule Subaccount (Joule + Build + AI services) and Connectivity Subaccount (SAP Connectivity Service)
- **Register systems in SAP BTP System Landscape** (auto-discovered if same contract; manually added otherwise)
- Configure BTP Role Collections for Joule and Joule Studio access
- Define LoB RBAC policies for Ariba, Concur, IBP, BN4L, Fieldglass, Commerce Cloud, Signavio, APM, and PAPM
- Set up S/4 RBAC for CFIN, IP, and IF systems
- Implement audit logging and telemetry infrastructure
- **Run Joule Booster** to create formations and configure destinations

L2 Deliverables (25%):

- Identity trust configuration documentation (including **SAML2/OIDC/SCIM** protocol specs and optional 3rd-party IdP setup)
- Role mapping matrix (including Work Zone CDM roles)
- Initial provisioning flows (IPS → Work Zone → LoB RBAC)
- **BTP subaccount topology documentation** (Joule + Connectivity subaccounts)
- **System Landscape registration procedures**
- **Work Zone configuration specifications** (entry point / launchpad role)
- **AI Core / AI Launchpad setup documentation**

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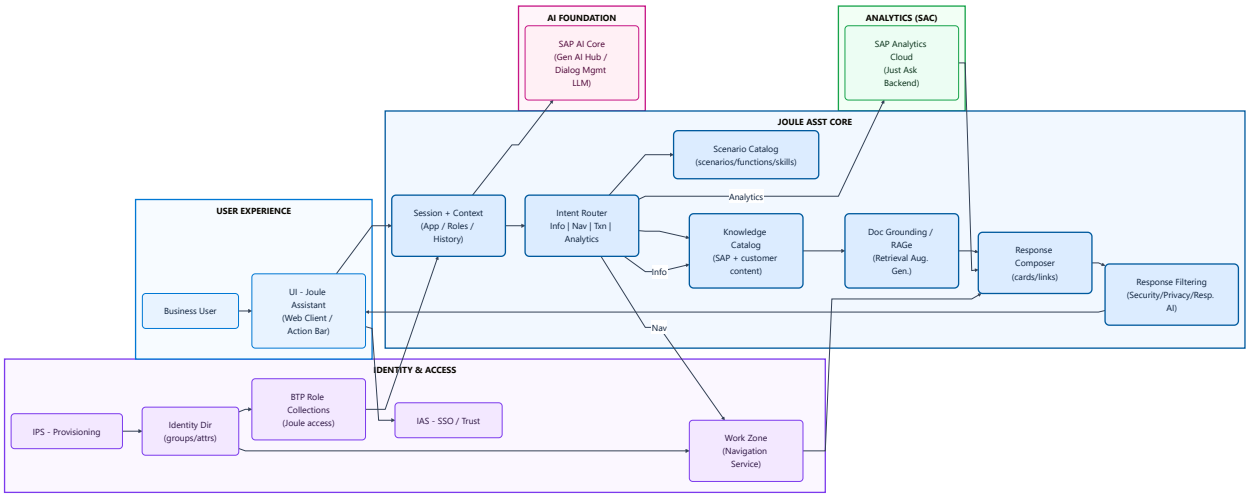
Phase 2 – Joule Assistant Core Enablement

Purpose: Enable the central Joule Assistant interface for informational queries, navigation, and analytics access.

Phase 2 – Joule Assistant Core Enablement Architecture



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Key Activities:

- Deploy Joule Assistant web client and action bar (via **SAP Build Work Zone** as entry point)
- Configure Intent Router for informational, navigation, and analytics queries
- **Populate Scenario Catalog** with metadata of available scenarios, functions, and skills across SAP cloud applications
- **Configure Knowledge Catalog** with SAP-owned content (SAP Help Pages) and customer-owned enterprise content
- Implement **Document Grounding with RAGe** (Retrieval Augmented Generation for enterprise) against Knowledge Catalog
- Configure Response Composer for cards, links, and confirmations
- **Implement Response Filtering** for enterprise security, data privacy, and responsible AI guardrails
- **Configure Dialog Management LLM** via SAP AI Core (Generative AI Hub) with partner LLM ecosystem
- **Integrate SAC Just Ask** for analytical insights (Joule SAC processes in backdrop)
- **Enable Work Zone Navigation Service** for cross-application navigation

L2 Deliverables (25%):

- Joule Assistant Core deployment architecture
- Intent routing configuration
- **Scenario Catalog population plan**
- **Knowledge Catalog content strategy** (SAP-owned + customer-owned sources)
- **RAGe / Document Grounding configuration**
- **Response Filtering and Responsible AI policy specifications**
- **SAC Just Ask integration specifications**
- **Work Zone navigation service configuration**
- **AI Core / Dialog Management LLM connection documentation**

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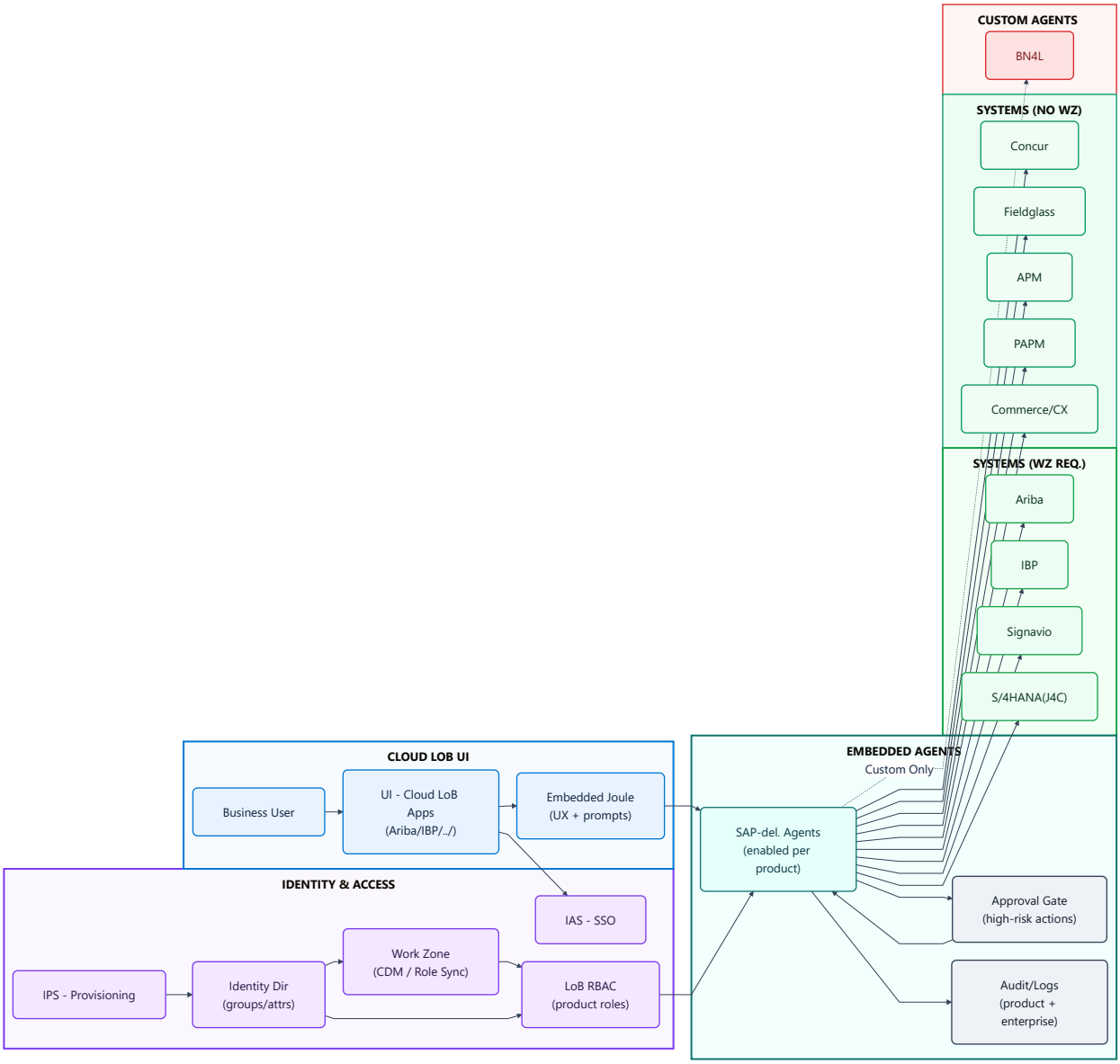
Phase 3 – Embedded Agents Activation

Purpose: Activate SAP-delivered embedded agents within Line of Business applications.

Phase 3 – Embedded Agents Features & Architecture



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Key Activities:

- Enable embedded Joule UX in each LoB application
- Activate SAP-delivered agents per product (Ariba, Concur, IBP, Fieldglass, Commerce, Signavio, APM, PAPM)
- Configure custom agents for BN4L (no SAP-delivered agents available)
- Configure LoB RBAC for product-specific roles
- **Configure Work Zone CDM content sync** for products requiring role synchronization
- Implement human-in-the-loop approval gates for high-risk actions
- Set up product and enterprise audit logging

L2 Deliverables (25%):

- Embedded agent activation matrix (with Work Zone requirements)
- LoB RBAC configuration
- **Work Zone CDM role sync specifications**
- Approval workflow specifications

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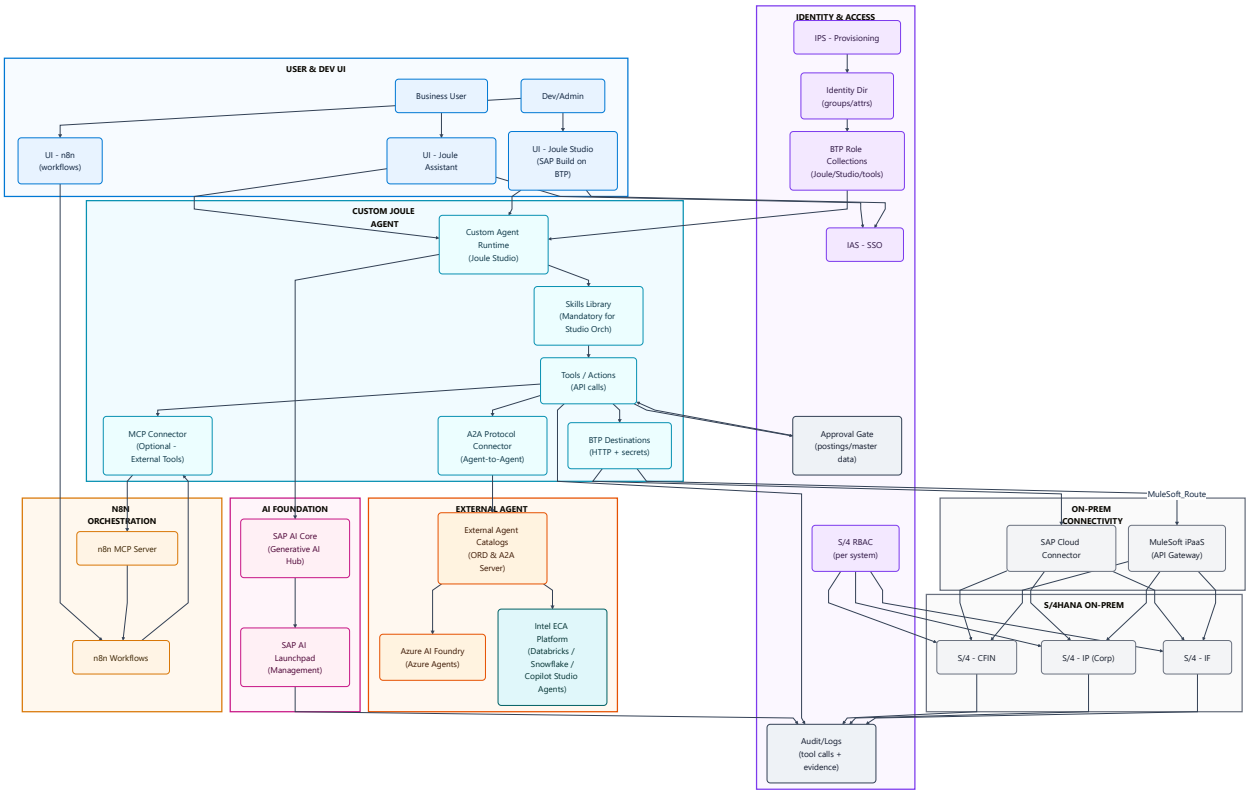
Phase 4 – Custom Agent Development

Purpose: Enable custom agent development via Joule Studio for S/4HANA on-prem integration and cross-system orchestration. **Note:** Skills Library is mandatory for Joule Studio orchestration; optional when using MCP/external tools.

Phase 4 – Custom Agent Development Architecture



Key Activities:

- Deploy Joule Studio (SAP Build on BTP) and **AI Agent** runtime
- Develop custom agent runtime with tools library
- **Configure SAP-delivered Skills** (pre-built Joule skills from SAP)
- **Develop Custom Skills** (customer-built skills for specific business processes)
- **Configure Skills Library** (mandatory for Studio orchestration)
- **Integrate SAP AI Core** for LLM processing and Generative AI Hub
- **Configure SAP Build Process Automation** for SAP-native workflow and process automation
- Configure **SAP Destination Service** with HTTP connections and secrets

- Set up **SAP Connectivity Service** (BTP-side connectivity proxy) and **SAP Cloud Connector** (on-prem agent) for on-prem connectivity
- Configure **MuleSoft iPaaS** as alternative on-prem connectivity option (API Gateway approach alongside SAP Cloud Connector)
- Implement MCP Tools Connector for optional external tools (Skills optional when using MCP)
- Configure **A2A Protocol Connector** for agent-to-agent interoperability with Intel ECA Platform agents (Databricks / Snowflake / Azure Copilot Studio) and Azure AI Foundry
- Configure n8n orchestration for workflow automation (optional / Intel-specific)
- Implement approval gates for postings and master data changes

L2 Deliverables (25%):

- Custom agent development standards
- **SAP-delivered Skills vs Custom Skills** inventory and configuration
- **Skills Library configuration** (Joule Studio flows)
- **AI Core integration specifications**
- **SAP Build Process Automation** workflow specifications
- SAP Destination Service configuration
- SAP Connectivity Service + Cloud Connector setup specifications
- **MuleSoft iPaaS** integration specifications (alternative on-prem connectivity)
- **A2A Protocol** configuration for ECA Platform agent interoperability
- **MCP connector documentation** (for external tool integration)

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Phase 5 – Governance & Operations

Purpose: Establish ongoing governance, monitoring, and operational excellence for the Joule implementation.

Key Activities:

- Implement comprehensive audit and telemetry collection
- Define human-in-the-loop approval policies for high-risk actions
- Establish agent lifecycle management processes
- Configure monitoring dashboards and alerting (via AI Launchpad)
- **Monitor Response Filtering and Responsible AI** compliance metrics
- Define incident response procedures
- Implement continuous improvement processes
- **Monitor AI Core usage** and LLM token consumption
- **Track Scenario Catalog and Knowledge Catalog** utilisation and coverage

L2 Deliverables (25%):

- Governance framework documentation
- Monitoring and alerting specifications (AI Launchpad integration)
- **Responsible AI compliance monitoring procedures**
- Operational runbooks
- **AI consumption tracking procedures**
- **Catalog coverage and utilisation reports**

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6. Joule Capability Matrix

> **How to read:** "Joule Assistant" = embedded conversational UX; "SAPâ€‘delivered Agents" = packaged LoB capabilities; "Skills" = **Required** (Studio) / **Optional** (MCP); "Custom Agents" = developer-built. **WZ Req** = Work Zone required for navigation/CDM role sync (not for custom agents alone).

Product	Joule Asst	SAP Agents	Skills	Custom Agents	WZ Reqâ□´	Dependencies
S/4HANA Cloud (J4C)	Yes	Yes	N/A	**No**[1]	Yes	Booster, IAS/IPS, WZ
SAP Analytics Cloud	Yes	Yes	Req/Opt	Yes	Yes	Work Zone, OAuth
SAP Ariba	Yes	Yes	Req/Opt	Yes	Yes	IAS, IPS → WZ
SAP Concur	Yes	Yes	Req/Opt	Yes	No	IAS, IPS
SAP IBP	Yes	Yes	Req/Opt	Yes	Yes	IAS, IPS → WZ
SAP Fieldglass	Yes	Yes	Req/Opt	Yes	Yes[5]	IAS, IPS, WZ
SAP Signavio	Yes	Yes	Req/Opt	Yes	Yes	IAS, IPS, Dest
SAP APM	Yes	Yes	Req/Opt	Yes	No	IAS, IPS
SAP PAPM	Yes	Yes	Req/Opt	Yes	No	IAS, IPS
SAP BN4L	No	**No**[2]	Req/Opt	**Yes**[2]	No	IAS, IPS, Custom Dev
S/4 On-Prem (CFIN/IF/IP)	No[3]	No	Req/Opt	**Yes**	No	CC, BTP Dest, S/4 RBAC
Commerce Cloud / CX	Yes	Partial	Req/Opt	Yes	No	IAS, IPS
SAP Build Code (J4D)	Yes	N/A	Required	Yes	No	Build Code, AI Core
Other Cloud Products	Varies	Varies	Req/Opt	Optional	Varies	IAS/IPS

Notes: [1]J4C is SAP-managed; use Side-by-Side extensions. [2]BN4L requires custom development (no SAP agents). [3]S/4 On-Prem uses custom agents via Cloud Connector. â□´**WZ Req** = Work Zone required for Joule Assistant navigation & CDM role sync; custom agent development alone does NOT require Work Zone. [5]Fieldglass requires WZ per SAP docs. **Skills:** Req=Required for Joule Studio; Opt=Optional for MCP/n8n. **PCE:** S/4HANA Private Cloud Edition (PCE) follows a similar pattern to J4C; consult SAP for PCE-specific Joule activation. **Other SAP Products:** SAP SuccessFactors, SAP Datasphere, SAP LeanIX, and SAP DSC are Joule-enabled in the SAP reference architecture but are not in current Intel scope.

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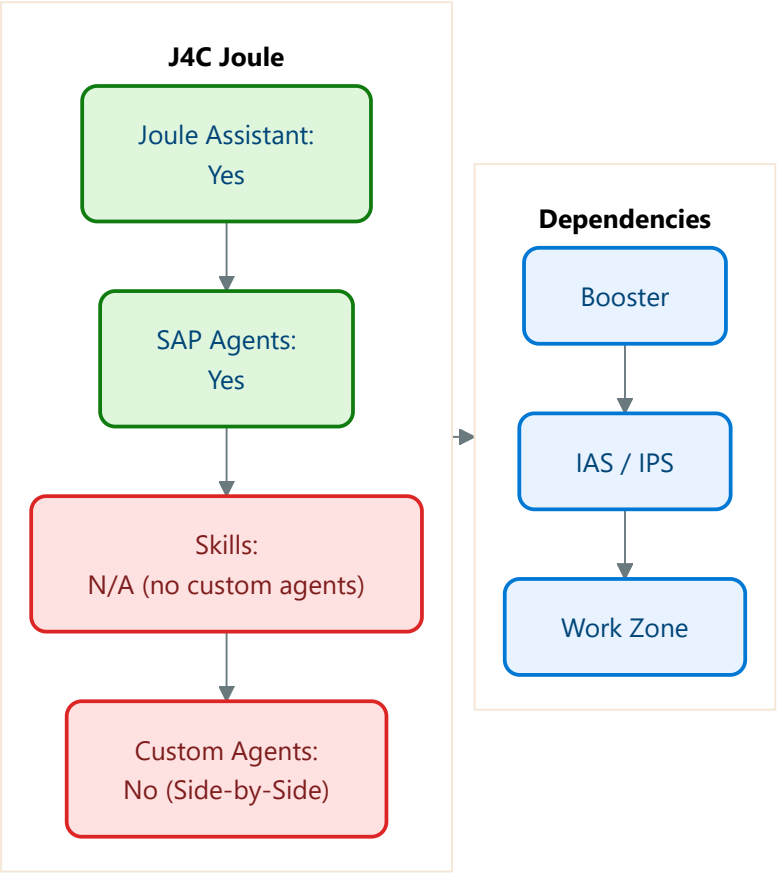
7. Product Capabilities

S/4HANA Cloud (J4C) – Joule Capabilities

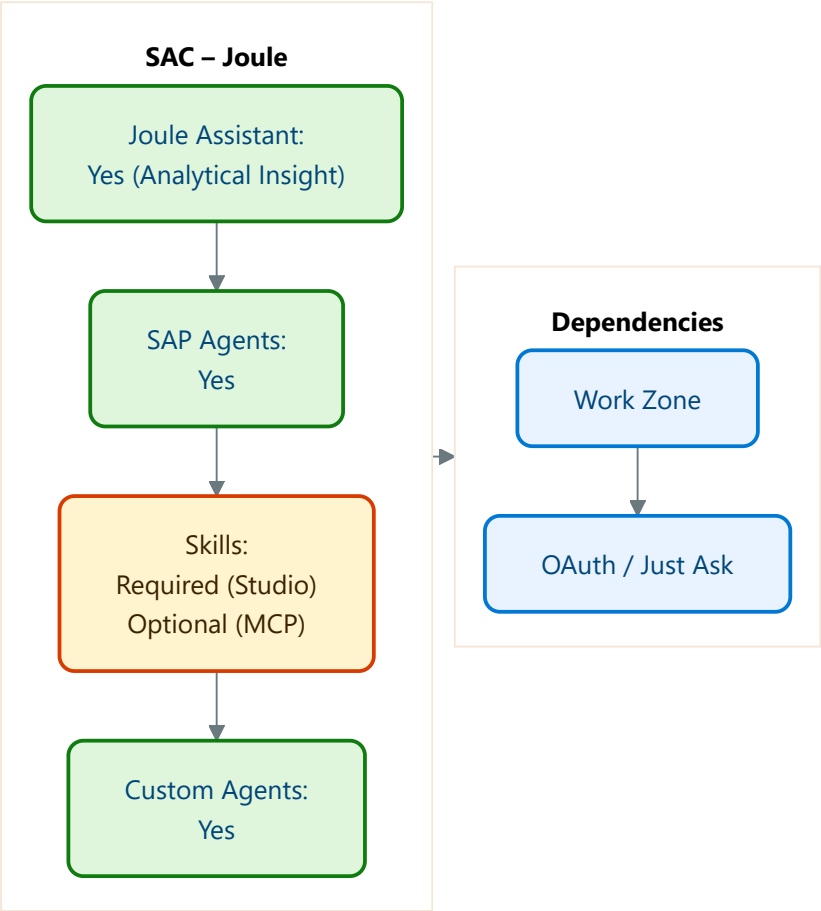
> **Note:** J4C is a SAP-managed offering. Custom agents cannot be built directly on J4C; use Side-by-Side extensions on BTP if custom development is needed.

Capability	Status	Dependencies
Joule Assistant	Yes	Booster, IAS/IPS, Work Zone
SAP-delivered Joule Agents	Yes	Booster, IAS/IPS, Work Zone
Skills (for Custom Agents)	N/A (no custom agents)	SAP-managed
Custom Joule Agents	**No** (Side-by-Side on BTP)	SAP Build / Studio



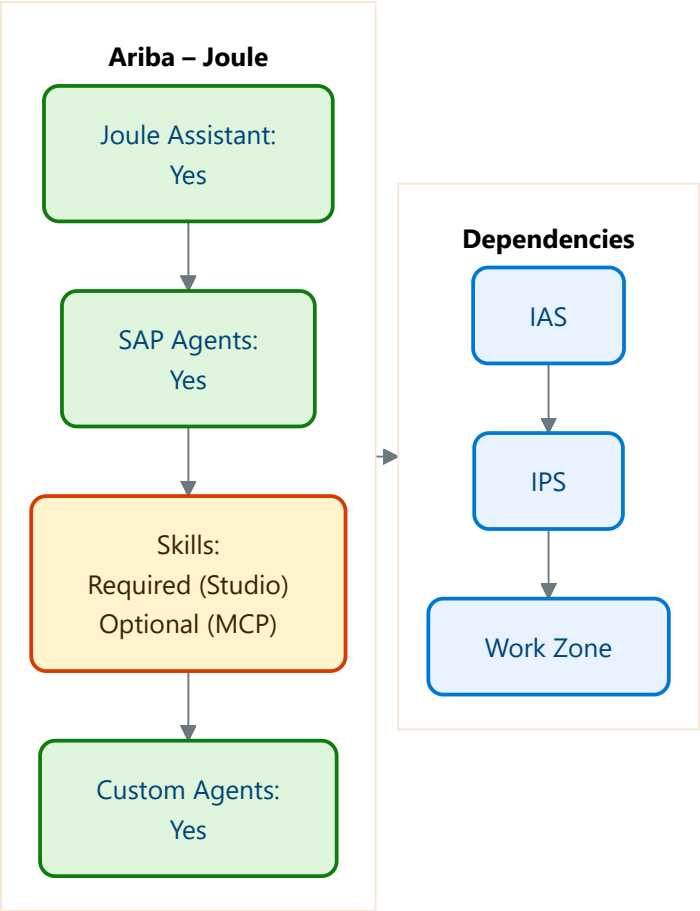
SAP Analytics Cloud (SAC) – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes (Analytical Insights)	Work Zone, OAuth clients
SAP-delivered Joule Agents	Yes	Work Zone, OAuth clients
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



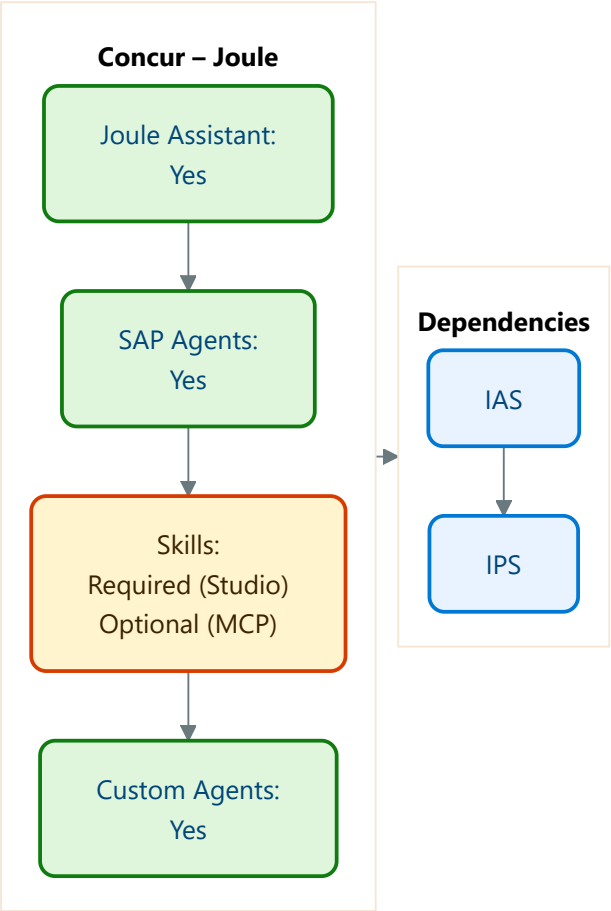
Ariba – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS, Work Zone (as needed)
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



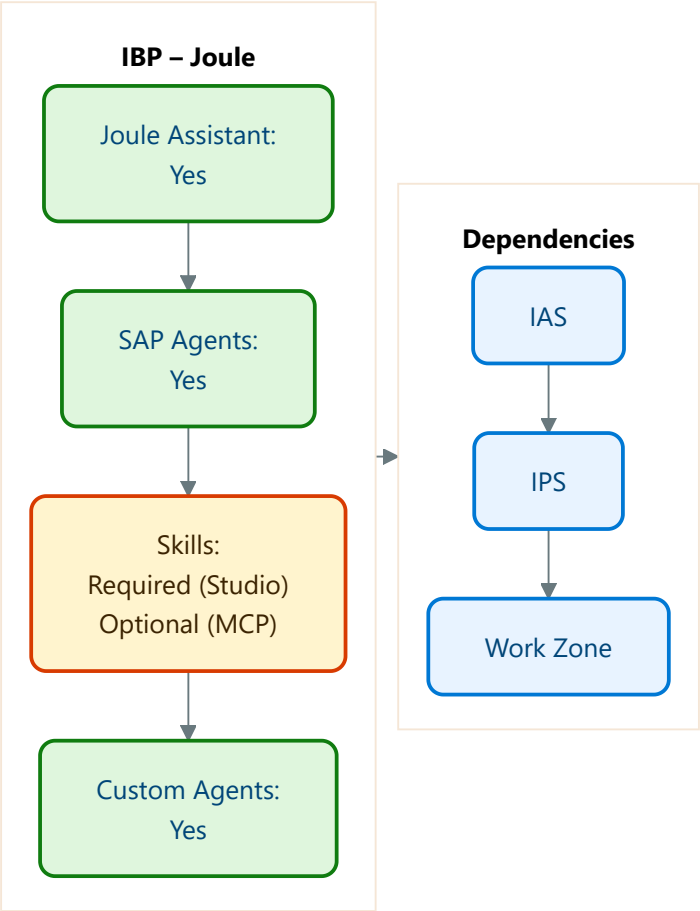
Concur – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



IBP – Joule Capabilities

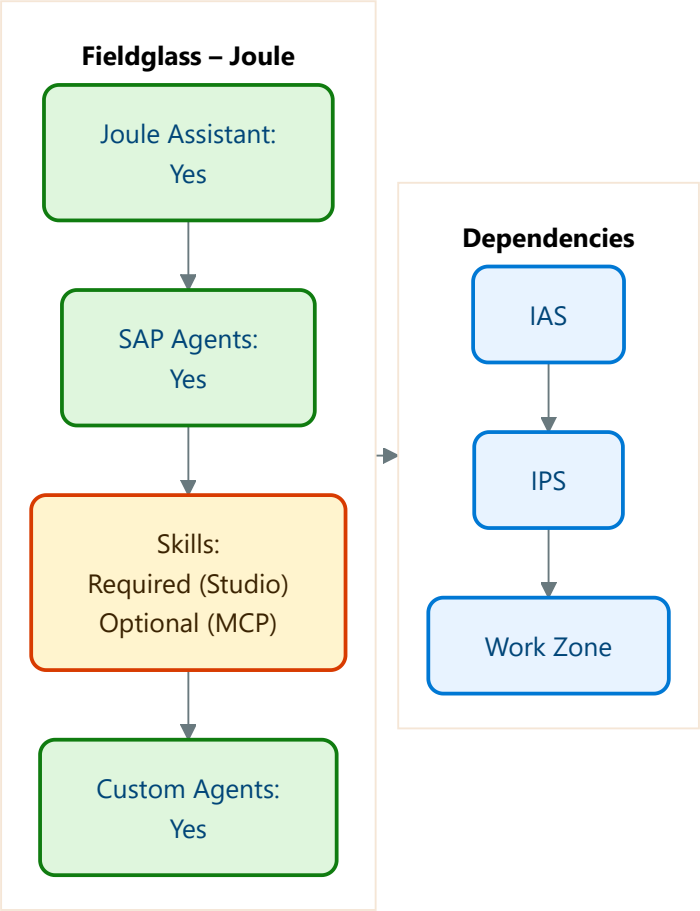
Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS, Work Zone
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



Fieldglass – Joule Capabilities

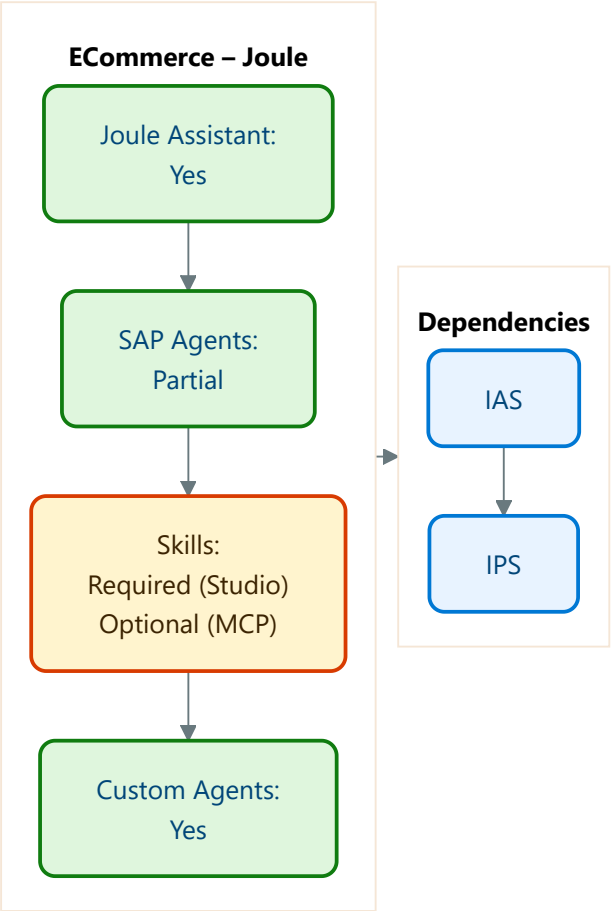
> **Note:** Fieldglass requires SAP Build Work Zone subscription per SAP documentation.

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS, Work Zone
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



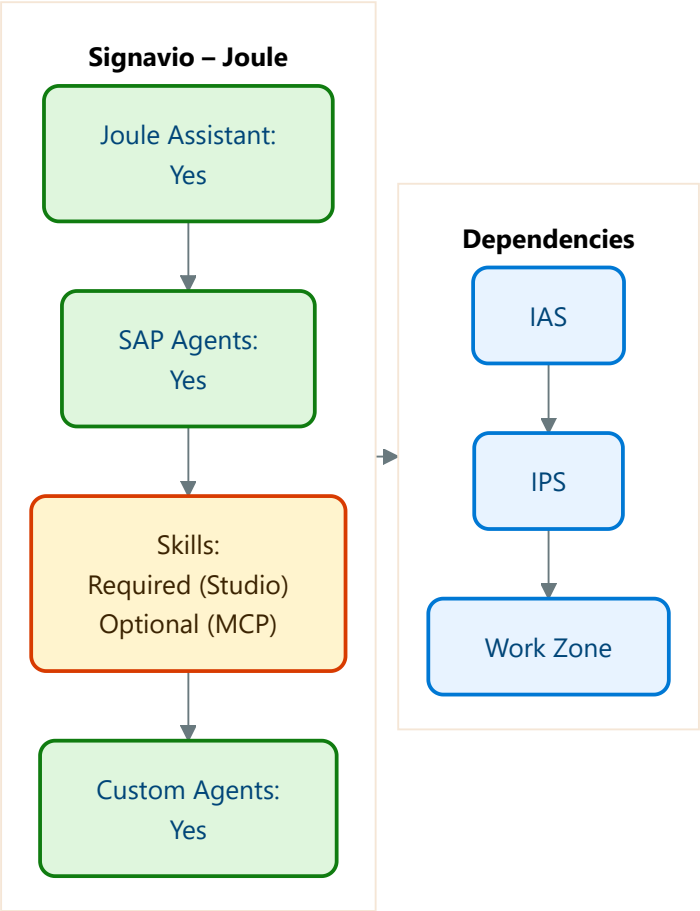
Commerce Cloud (CX) – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS
SAP-delivered Joule Agents	Partial	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



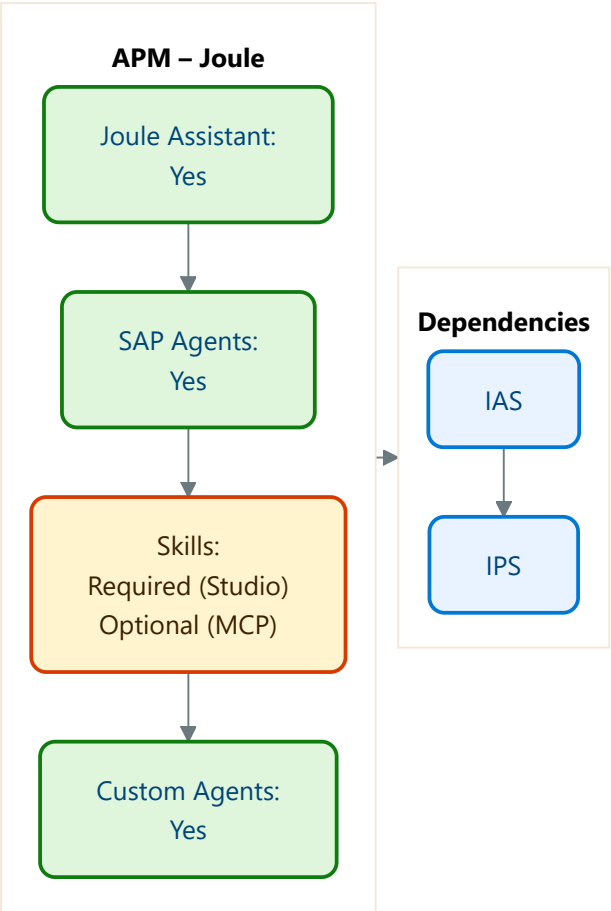
Signavio – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS, Work Zone
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



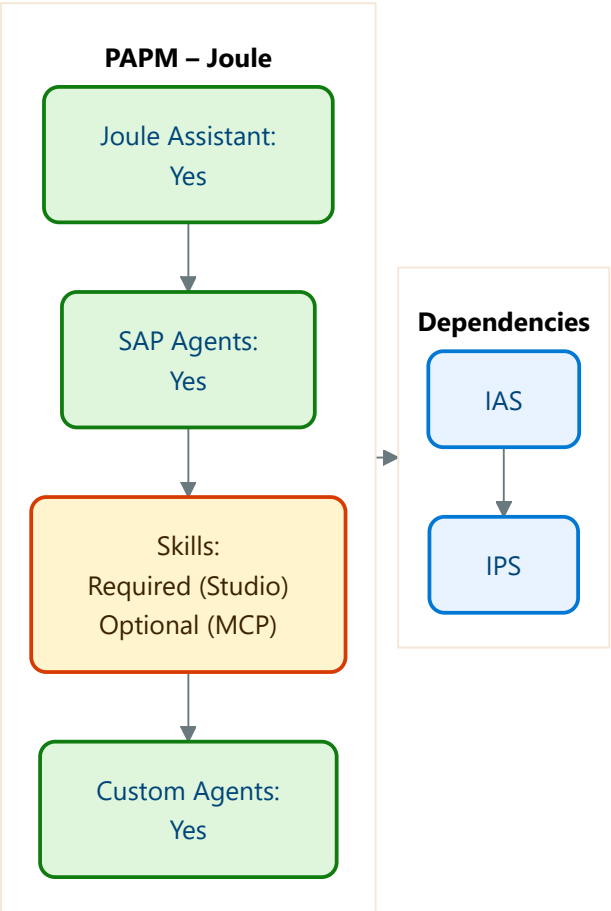
APM (Asset Performance Management) – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



PAPM (Profitability & Performance Mgmt) – Joule Capabilities

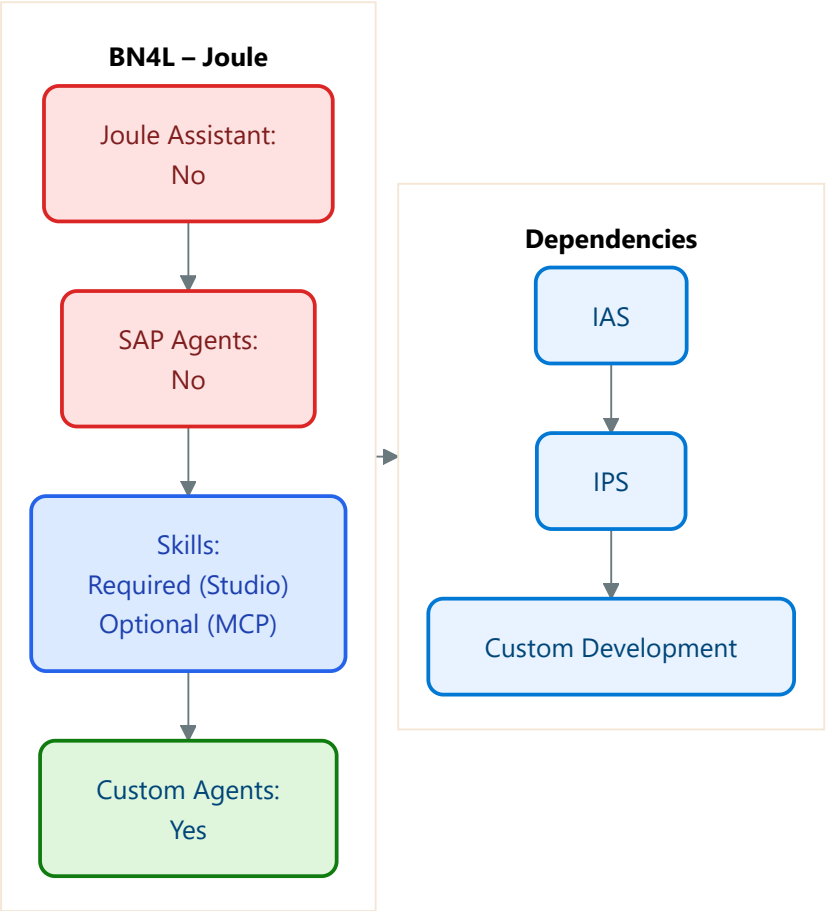
Capability	Status	Dependencies
Joule Assistant	Yes	IAS, IPS
SAP-delivered Joule Agents	Yes	IAS, IPS
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio



BN4L (Business Network for Logistics) – Joule Capabilities

> **Important:** BN4L does not support SAP-delivered agents or embedded Joule Assistant. Only custom agents can be built via Joule Studio.

Capability	Status	Dependencies
Joule Assistant	No	N/A
SAP-delivered Joule Agents	**No**	N/A
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	**Yes**	SAP Build / Studio, Custom Development

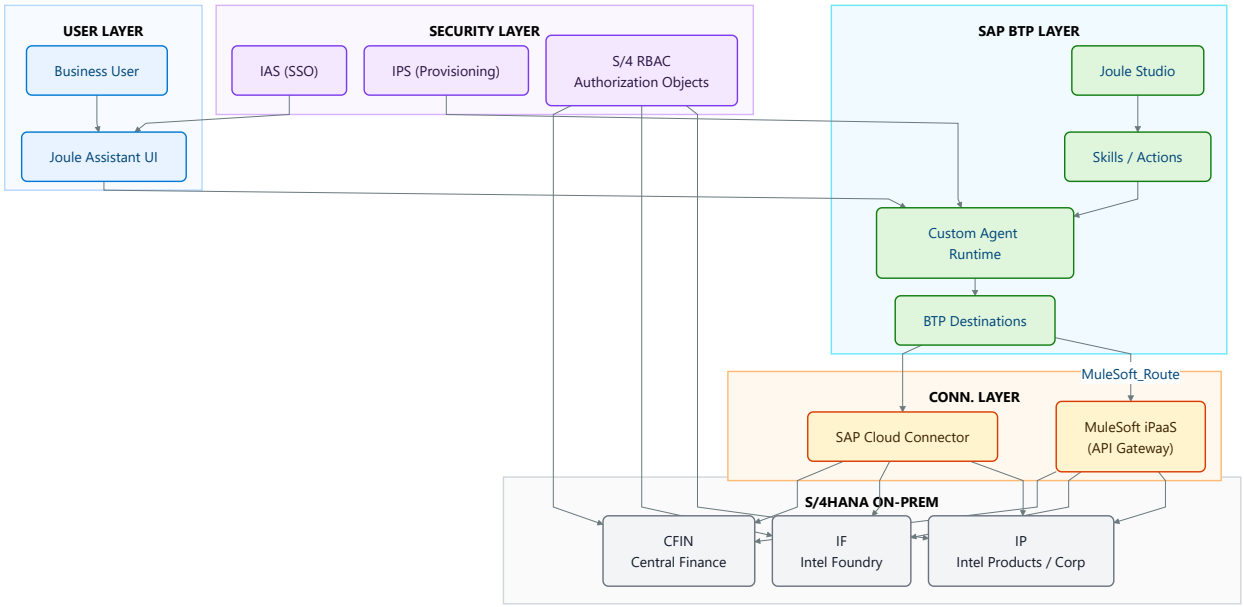


S/4HANA On-Premise – Joule Capabilities

> **Context:** Intel operates S/4HANA on-premise systems for Central Finance (CFIN), Intel Foundry (IF), and Intel Products / Corp (IP / Corp). These systems connect to SAP BTP via Cloud Connector for custom agent development.

Capability	Status	Dependencies
Joule Assistant	No (embedded)	Cloud Connector, BTP
SAP-delivered Joule Agents	No	N/A
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio, BTP Destinations
Custom Joule Agents	**Yes**	Cloud Connector, S/4 RBAC, BTP

Architecture Pattern: S/4HANA On-Premise to Joule



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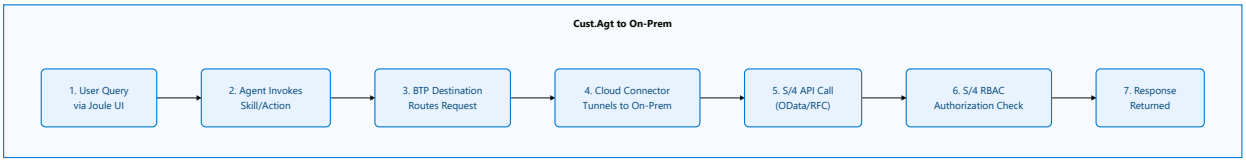
Process Flow: Custom Agent Accessing S/4HANA On-Premise



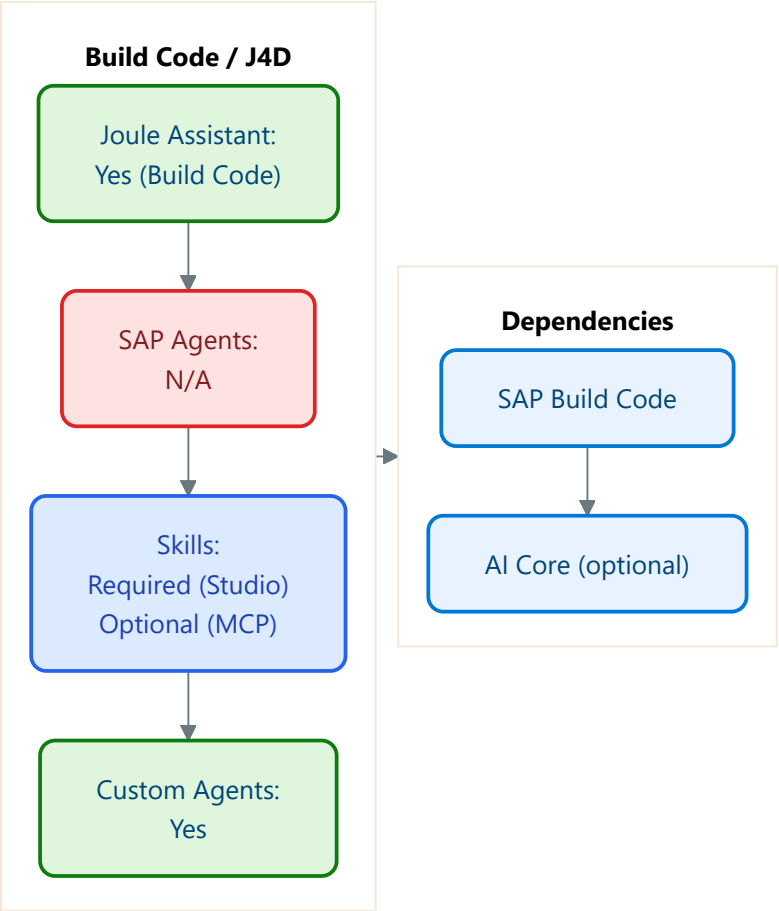
Key Implementation Requirements

Requirement	Description
Cloud Connector	Deploy and configure SAP Cloud Connector for secure on-premise connectivity
BTP Destinations	Create HTTP destinations pointing to S/4HANA systems (CFIN, IF, IP)
S/4 RBAC	Configure authorization objects for API access (e.g., BAPI, OData services)
Skills Definition	Define Skills in Joule Studio that map to S/4HANA API endpoints
Principal Propagation	Optional: Enable SSO from BTP to S/4 via principal propagation

SAP Build Code / ABAP Cloud (J4D) – Joule Capabilities

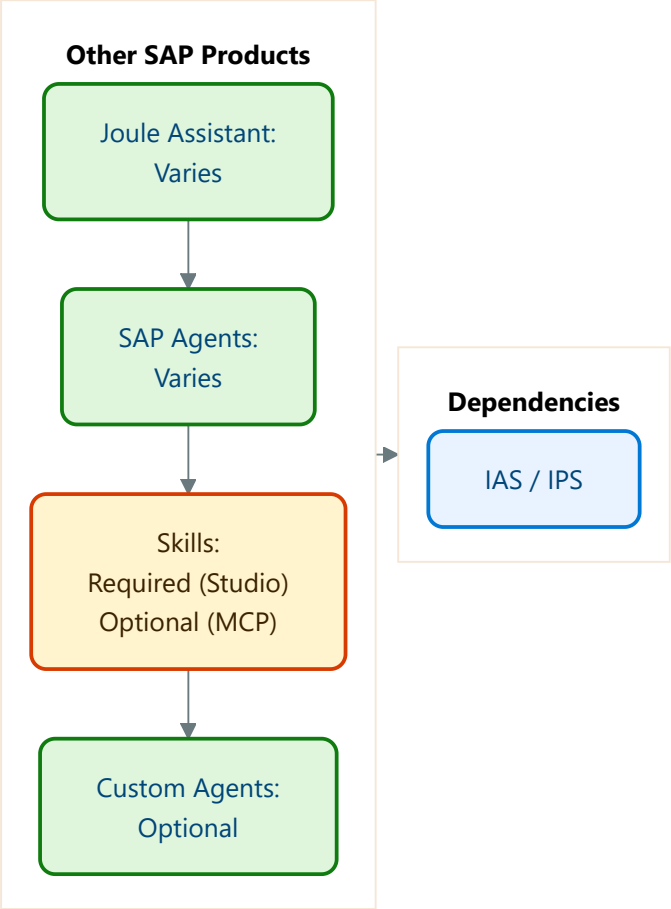
> **Note:** SAP Build Code includes Joule as an embedded AI Assistant for code generation, data models, services, and sample data. Joule in Build Code assists developers with CAP, Fiori, SAPUI5, and mobile development.

Capability	Status	Dependencies
Joule Assistant	**Yes** (in Build Code)	SAP Build Code
SAP-delivered Joule Agents	N/A	N/A
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Yes	SAP Build / Studio, AI Core (optional)


Build Code / J4D
Joule Assistant:
Yes (Build Code)
SAP Agents:
N/A
Skills:
Required (Studio)
Optional (MCP)
Custom Agents:
Yes
Dependencies
SAP Build Code
AI Core (optional)

Other Cloud Products – Joule Capabilities

Capability	Status	Dependencies
Joule Assistant	Varies	IAS/IPS as applicable
SAP-delivered Joule Agents	Varies	IAS/IPS as applicable
Skills (for Custom Agents)	**Required** (Studio) / Optional (MCP)	SAP Build / Studio
Custom Joule Agents	Optional	SAP Build / Studio



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Skills Architecture: Joule Studio vs MCP

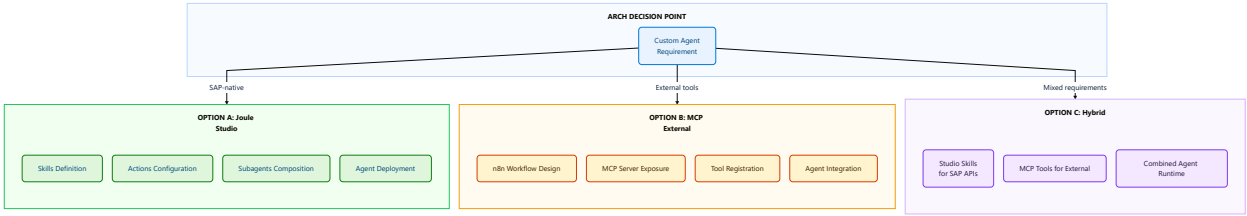
> **Purpose:** This section explains when Skills are **Required** vs **Optional** based on the orchestration architecture chosen for building custom Joule agents.

Understanding Skills in Joule Architecture

Skills are reusable action definitions that enable Joule agents to perform specific tasks (API calls, data retrieval, business actions). The requirement for Skills depends on your orchestration approach:

Orchestration Approach	Skills Requirement	When to Use
Joule Studio	**Required**	Building agents entirely within SAP ecosystem using Studio visual tools
MCP (Model Context Protocol)	**Optional**	Using external orchestration tools (n8n, LangChain) that expose tools via MCP
Hybrid	**Partial**	Some actions via Studio Skills, others via MCP external tools

Architecture Decision: Joule Studio vs MCP



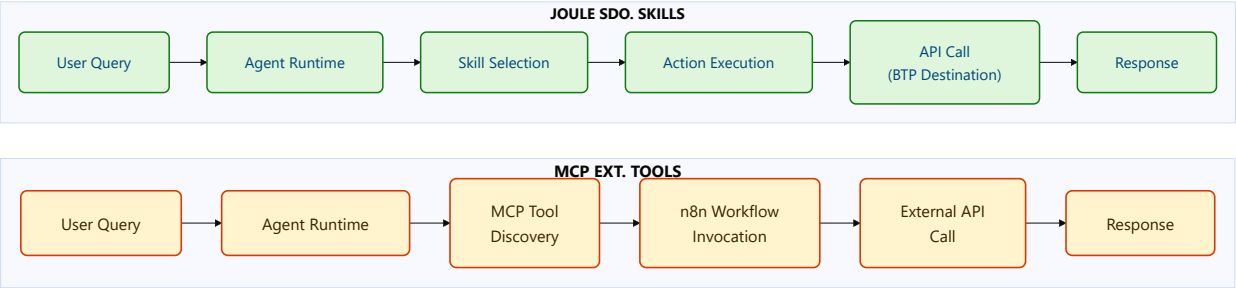
When to Choose Each Approach

Approach	Best For	Considerations
Joule Studio	SAP-centric processes, standard LoB integrations	Full SAP governance, native monitoring, simplified deployment
MCP with n8n	Complex workflows, non-SAP integrations, retry logic	Requires MCP server setup, external tool management
Hybrid	Mixed SAP and external systems	Combines benefits, requires coordination between tools

Skills Requirement by Product

Product	Default Approach	Skills Status	Notes
S/4HANA On-Prem (CFIN, IF, IP)	Studio	**Required**	Skills define S/4 API calls via BTP Destinations
SAP Build Code / J4D	Studio	**Required**	Developer builds Skills for custom agents
Ariba, Concur, IBP, Fieldglass	Studio or MCP	**Required (Studio) / Optional (MCP)**	Choice depends on integration complexity
SAC	MCP preferred	**Optional**	Just Ask backend handles most queries
APM, PAPM	MCP preferred	**Optional**	Standard analytics-focused interactions
BN4L	Studio	**Required**	No SAP-delivered agents; custom only

Process Flow: Skills-Based Agent Execution



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8. Platform Stage Lanes

Platform Stage Lanes – Simplified View

This simplified lane view provides an executive-friendly overview of the Joule deployment stages across the SAP landscape.

Stage	Components	Responsibility	SLA
Identity Foundation	IAS, IPS, Identity Directory, Work Zone	SAP Basis / Identity Team	99.9% availability
AI Infrastructure	SAP AI Core, AI Launchpad, Generative AI Hub	AI Platform Team	99.5% availability
Joule Assistant Core	Scenario Catalog, Knowledge Catalog, RAGe, Response Filtering	SAP Application Team	99.5% availability
Embedded Agents	LoB-specific agents (Ariba, Concur, IBP, etc.)	LoB Application Teams	99.5% availability
Custom Agents	Joule Studio, Skills Library, Tools, BTP Destinations	Custom Development Team	99.0% availability
Connectivity	Cloud Connector, Connectivity Service, Destination Service, MuleSoft	Integration Team	99.5% availability
Governance	Audit logging, HITL approval gates, AI Launchpad monitoring	Governance / Compliance	99.9% audit capture

Deployment Progression

```
Stage 1 [==========] Complete – Identity Foundation (IAS/IPS/Work Zone)
Stage 2 [========  ] 80% – AI Infrastructure (AI Core/Launchpad)
Stage 3 [======    ] 60% – Joule Assistant Core (Scenario/Knowledge Catalogs)
Stage 4 [====      ] 40% – Embedded Agents (LoB Activation)
Stage 5 [==        ] 20% – Custom Agents (Joule Studio/Skills)
Stage 6 [=         ] 10% – Governance & Operations
```

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9. TOGAF BDAT Crosswalk

TOGAF BDAT Crosswalk – Conceptual Mapping

This conceptual mapping aligns the SAP Joule architecture to TOGAF Business, Data, Application, and Technology (BDAT) domains, supporting enterprise architecture governance.

Business Architecture (B)

Capability	Description	Value Delivered
SAP Transaction Acceleration	Natural language interaction with SAP business processes	Reduced task completion time
Cross-LoB Agent Assistance	Embedded agents across Ariba, Concur, IBP, and more	Consistent AI experience across SAP portfolio
Knowledge-Driven Decisions	RAGe-powered enterprise knowledge retrieval	Faster, informed decision-making
Analytics Self-Service	SAC Just Ask for natural language business insights	Democratized analytics access

Data Architecture (D)

Layer	Purpose	Quality Standard
Knowledge Catalog	SAP-owned and customer-owned knowledge base	Curated, version-controlled
Scenario Catalog	Metadata registry of Joule scenarios and skills	Registered, governed
Document Grounding	RAGe-powered retrieval for informational queries	Chunked, embedded, indexed
Identity Directory	User attributes, groups, team memberships	Authoritative, SCIM-synced

Application Architecture (A)

Application	Role	Integration Pattern
SAP Build Work Zone	Entry point, navigation, CDM role sync	IPS SCIM provisioning
SAP AI Core	LLM runtime, Generative AI Hub	Dialog Management LLM
Joule Studio	Custom agent development	Skills + MCP + A2A
Scenario Catalog	Scenario/skill registry	REST API lookup
Knowledge Catalog	Knowledge base for RAGe	Document grounding
SAP Cloud Connector	On-prem connectivity	Secure tunnel to S/4HANA
SAP Destination Service	Route management	HTTP + credential management

Technology Architecture (T)

Technology	Purpose	Provider
SAP BTP	Cloud platform for extensions	SAP
SAP AI Core	AI/ML runtime	SAP
SAP Cloud Identity Services	Identity management	SAP
SAP Cloud Connector	On-prem tunnel	SAP
MuleSoft iPaaS	Alternative on-prem API gateway	MuleSoft / Salesforce
n8n	Workflow automation	n8n (open source)

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10. Ownership & Operating Model

Ownership / Operating Model

This view maps Joule platform components to responsible teams, supporting RACI discussions and operational governance.

RACI Matrix

Component	SAP Basis	Identity Team	LoB App Teams	Custom Dev Team	AI Platform	Governance
IAS / IPS / Identity Directory	C	**R/A**	I	I	I	I
SAP Build Work Zone	**R/A**	C	I	I	I	I
SAP AI Core / AI Launchpad	C	I	I	C	**R/A**	I
Scenario Catalog	C	I	**R/A**	C	C	I
Knowledge Catalog	I	I	**R/A**	C	C	I
SAP-delivered Agents (LoB)	I	I	**R/A**	I	C	I
Custom Agents (Joule Studio)	I	I	C	**R/A**	C	I
Skills Library	I	I	C	**R/A**	C	I
Cloud Connector / Connectivity	**R/A**	I	I	C	I	I
BTP Destinations	**R/A**	I	I	C	I	I
Response Filtering / Responsible AI	I	I	I	I	C	**R/A**
Audit Logging / Telemetry	I	I	I	I	C	**R/A**
HITL Approval Gates	I	I	C	C	I	**R/A**

RACI Legend:

Code	Role	Definition
R/A	Responsible & Accountable	Owns delivery and final decision authority
C	Consulted	Provides input before decisions are made
I	Informed	Notified after decisions are made

Support Model

Tier	Scope	Team	SLA
L1	User issues, Joule access requests	Service Desk	4 hours
L2	LoB agent issues, Scenario Catalog gaps	LoB Application Teams	8 hours
L3	AI Core issues, custom agent failures	AI Platform / Custom Dev	24 hours
L4	Architecture, design decisions, governance	Enterprise Architecture	5 days

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11. L2 Architecture Deliverables

Identity & Access Architecture

- Configure **SAP Cloud Identity Services** (IAS + IPS + Identity Directory) as unified identity foundation with common instance (prod + non-prod) using single domain URL (`cloud.sap.com`)
- Configure IAS as central identity provider with SSO via **SAML2/OIDC** trust relationships across all SAP applications
- Optionally integrate **3rd-party Identity Providers** via SAML trust with SAP Cloud Identity Services
- Implement IPS for automated **SCIM-based** user provisioning, group synchronization, and attribute mapping
- Establish Identity Directory as authoritative source for user attributes, groups, and team memberships
- Define BTP Role Collections for Joule Core and Joule Studio access control
- Configure LoB-specific RBAC policies for Ariba, Concur, IBP, BN4L, Fieldglass, Commerce Cloud, Signavio, APM, and PAPM
- Implement S/4 RBAC for CFIN, IP, and IF systems with appropriate authorization objects
- **Document BTP subaccount topology** (Joule Subaccount + Connectivity Subaccount)

Integration Architecture

- Configure **SAP Destination Service** for route management with secure credential management
- Configure **SAP Connectivity Service** (BTP connectivity proxy) for hybrid landscape routing
- Implement **SAP Cloud Connector** (on-prem agent) for secure on-premises connectivity to S/4HANA systems
- **Register systems in SAP BTP System Landscape** (auto-discovered if same contract; manually added otherwise)
- Define API gateway patterns for external system integration
- Establish MCP Tools Connector configuration for optional external tool integration
- **Configure SAP Build Process Automation** for SAP-native workflow orchestration

- Configure n8n orchestration patterns for workflow automation (optional / Intel-specific)
- Implement event-driven integration patterns where applicable

Security Architecture

- Implement human-in-the-loop approval gates for high-risk actions (postings, master data changes)
- **Configure Response Filtering** for enterprise security, data privacy, and responsible AI compliance before response delivery
- **Define Responsible AI policies** aligned with SAP AI ethics policy and UNESCO's 10 guiding principles on Ethics of AI
- Configure comprehensive audit logging and telemetry collection across all Joule interactions
- Define data classification and handling policies for AI-processed information
- Establish secure credential management via SAP Destination Service and secrets
- Implement network security controls for Cloud Connector / Connectivity Service communications
- Define incident response procedures for AI-related security events

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11. L2 Architecture Deliverables...continued

Agent Architecture

- Define custom agent development standards and best practices for Joule Studio and **AI Agent** runtime
- Establish tools library structure with **SAP-delivered Skills, Custom Skills**, actions, and subagent definitions
- **Document Scenario Catalog** population strategy and Knowledge Catalog content curation processes
- Configure agent lifecycle management processes (development, testing, deployment, retirement)
- Implement agent versioning and rollback capabilities
- Define agent monitoring and performance metrics
- Establish agent governance policies and approval workflows

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12. Governance & Operations

Governance Framework

AI Governance

Domain	Owner	Controls
Model Approval	AI Platform Team	SAP AI Core model registry, Generative AI Hub
Prompt Safety	AI Platform Team	Response Filtering, Responsible AI guardrails
Action Approval	Business Owners	HITL for high-risk actions (postings, master data changes)
Audit & Evidence	Compliance	Comprehensive logging via AI Launchpad telemetry
Responsible AI	Governance	Aligned with SAP AI ethics policy and UNESCO's 10 guiding principles

Data Governance

Domain	Owner	Controls
Knowledge Catalog	LoB Application Teams	Content curation, version control, freshness SLAs
Scenario Catalog	LoB Application Teams	Scenario registration, skill governance
Identity Data	Identity Team	SCIM provisioning, group sync, attribute management
Data Classification	Governance	Classification-based access controls for AI-processed data

Operational Procedures

Monitoring & Alerting

Metric	Threshold	Action
Joule response error rate	> 2%	Alert AI Platform Team
AI Core LLM latency P95	> 5 sec	Investigate model performance
Agent approval backlog	> 25 pending	Escalate to business owners
Response Filtering blocks	> 5% of requests	Review Responsible AI policies
Token consumption	> 80% budget	Alert for capacity planning

Incident Management

Severity	Definition	Response Time	Resolution Time
P1	Joule platform outage	15 min	4 hours
P2	LoB agent failure (production)	1 hour	8 hours
P3	Custom agent issue	4 hours	24 hours
P4	Enhancement request	1 week	Sprint planning

Change Management

Change Type	Approval	Lead Time
LoB agent activation	LoB Application Team	5 days
Custom agent deployment	AI Platform + Governance	10 days
Skills Library update	Custom Dev Team	3 days
Emergency (security)	Emergency CAB	2 hours

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13. Architectural Flows

Identity & Access Flow (Foundation)

```
User → SAP Cloud Identity Services:
  IAS (SSO via SAML2/OIDC) → [Optional: 3rd-Party IdP (SAML Trust)] →
  IPS (SCIM Provisioning) → Identity Directory (Groups/Attributes) →
  Work Zone (CDM/Role Sync) → LoB RBAC (Product-Specific Roles)

Note: SAP recommends a common IAS instance (prod + non-prod) with
single domain URL (cloud.sap.com). BTP uses two subaccounts:
Joule Subaccount and Connectivity Subaccount.
```

Informational Flow

```
User → Joule Assistant UI → IAS (SSO) → Work Zone (Entry Point) →
Joule Session → Intent Router (Info) → Knowledge Catalog →
Document Grounding / RAGe (Retrieval Augmented Generation) →
Dialog Management LLM (AI Core) → Response Composer →
Response Filtering (Security/Privacy/Responsible AI) → User

Note: Knowledge Catalog contains SAP-owned content (SAP Help Pages)
and customer-owned content. RAGe grounds LLM responses on
retrieved enterprise documents.
```

Analytics Flow (SAC Just Ask)

```
User → Joule Assistant UI → IAS (SSO) → Joule Session →
Intent Router (Analytics) → Scenario Catalog →
SAP AI Core (Dialog Management LLM) →
SAC Just Ask (Backend Processing) → Response Composer →
Response Filtering (Security/Privacy/Responsible AI) → User

Note: SAC Just Ask is the backend engine; Joule operates in the backdrop
processing natural language analytical queries.
```

Navigation Flow

```
User → Joule Assistant UI → Intent Router (Navigation) →
Scenario Catalog → Work Zone (Navigation Service) →
Target Application → User
```

Embedded Transaction Flow (Work Zone Required Systems)

```
User → LoB UI (Ariba/IBP/Signavio/J4C/PCE) → IAS (SSO) →
IPS → Work Zone (Role Sync) → Embedded Joule →
Scenario Catalog → SAP-delivered Agent → LoB RBAC → System of Record →
Approval Gate (if required) → Response Filtering → Audit Log → User
```

Embedded Transaction Flow (No Work Zone)

```
User → LoB UI (Concur/Fieldglass/APM/PAPM/CX) → IAS (SSO) →
Direct LoB RBAC → Embedded Joule → Scenario Catalog →
SAP-delivered Agent → System of Record →
Approval Gate (if required) → Response Filtering → Audit Log → User
```

Custom Transaction Flow (Joule Studio – Skills Required)

```
User → Joule Assistant UI → IAS (SSO) → Joule Session →
Intent Router → Scenario Catalog → Custom Agent Runtime →
SAP-delivered Skills + Custom Skills → Tools Library →
SAP AI Core (Dialog Management LLM) →
SAP Destination Service → SAP Connectivity Service →
Cloud Connector → S/4HANA On-Prem →
Approval Gate (if required) → Response Filtering → Audit Log → User
```

Custom Transaction Flow (MCP/External Tools – Skills Optional)

```
User → Joule Assistant UI → IAS (SSO) → Custom Agent Runtime →
MCP Connector → External Tools (n8n/3rd Party) → SAP AI Core (LLM) →
SAP Destination Service → Target Systems →
Approval Gate (if required) → Response Filtering → Audit Log → User

Note: Skills Library is optional when using MCP/external orchestration (n8n, etc.).
```

BN4L Custom Agent Flow

```
User → Joule Assistant UI → IAS (SSO) → Custom Agent Runtime →
Custom Skills (Required) → Tools Library → SAP AI Core (LLM) →
SAP Destination Service → BN4L →
Approval Gate (if required) → Response Filtering → Audit Log → User

Note: BN4L only supports custom agents; no SAP-delivered agents available.
```

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14. Related Architecture Patterns

SAP Joule is one of **three complementary AI architecture patterns** at Intel. Each pattern is purpose-built for a distinct class of use cases and maintains its own runtime, identity model, and governance boundaries.

Pattern	Scope	Entry Point
SAP Joule (this document)	SAP transactions, navigation, analytics, SAP knowledge	Joule Assistant bar in SAP apps
ECA / Azure Copilot Studio	Enterprise data analytics, agentic workflows, non-SAP integrations	Copilot Agent UX (Teams / MS365)
iGPT Platform	General-purpose chat, custom assistants, marketplace, document Q&A	igpt.intel.com/chat

High-Level Integration Points

- **Joule → ECA/Azure**: SAP data replicated via SLT/CIF is queryable in ECA through governed MCP tools. Hybrid use cases (e.g., SAP approval followed by ECA analytics) leverage MCP bridge and A2A protocol.
- **Joule → iGPT**: No direct integration. General-purpose AI needs are served by iGPT as a standalone platform.
- **BDC Connect**: SAP Business Data Cloud enables zero-copy data sharing between Joule and ECA/Azure runtimes.

Selection Guidance

For detailed use case routing, decision frameworks, and guidance on choosing between the three patterns, refer to:

> 📄 **AI Architecture Selection Guide** — `AI-Architecture-Selection-Guide.md`

Related Documents

Document	Location
ECA & AI Enablement Architecture	`ECA_AI_Enablement_Architecture_DocumentV1.md`
iGPT Architecture Overview	`Current AI Architecture Options/iGPT-Architecture-Overview.md`
AI Architecture Selection Guide	`AI-Architecture-Selection-Guide.md`

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Appendix A: Component Glossary

Abbr.	Full Name	Description
ABAP	Advanced Business Application Programming	SAP proprietary programming language
APM	Asset Performance Management	Asset lifecycle management solution
BN4L	Business Network for Logistics	SAP logistics network solution
BTP	Business Technology Platform	Cloud platform for extensions & integrations
CDM	Content Deployment Manager	Work Zone content sync service
CFIN	Central Finance	S/4HANA central finance solution
CX	Customer Experience	SAP Commerce Cloud suite
FG	Fieldglass	Vendor management system
HITL	Human-in-the-Loop	Approval requiring human intervention
IAS	Identity Authentication Service	Cloud identity auth (SAML2/OIDC)
IBP	Integrated Business Planning	Supply chain planning solution
IF	Intel Foundry	S/4HANA on-prem (Foundry ops)
IP	Intel Corp / Products	S/4HANA on-prem (Corp/Products ops)
IPS	Identity Provisioning Service	Identity sync service (SCIM-based)
J4C	Joule for Consultants	Joule capabilities for S/4HANA Cloud
J4D	Joule for Developers	Joule for Build Code / ABAP Cloud
Knowledge Catalog	Knowledge Catalog	Knowledge registry for RAGe grounding
LoB	Line of Business	SAP business application category
MCP	Model Context Protocol	Standard for tool/workflow integration
n8n	n8n	Open-source workflow automation platform
OIDC	OpenID Connect	Auth protocol for Joule â†" Cloud Identity
PAPM	Profitability & Performance Mgmt	SAP analytics solution
PCE	Private Cloud Edition	S/4HANA Private Cloud deployment
RAGe	Retrieval Augmented Generation (enterprise)	Enterprise RAG for grounding LLM responses
RBAC	Role-Based Access Control	Permission management by role
SAC	SAP Analytics Cloud	Cloud analytics platform
SAML2	Security Assertion Markup Language 2.0	SSO protocol for BTP â†" identity providers
Scenario Catalog	Scenario Catalog	Registry of Joule scenarios & skills
SCIM	System for Cross-domain Identity Mgmt	Automated identity provisioning protocol
Signavio	SAP Signavio	Process management & mining solution
SoD	Segregation of Duties	Control preventing conflict of interest
SSO	Single Sign-On	Unified authentication mechanism

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Document Control

Version	Date	Author	Changes
1.0	January 2026	Sajiv Francis, Don Meyers	Initial release
1.1	February 2026	Sajiv Francis	Aligned with SAP Reference Architecture (Scenario Catalog, Knowledge Catalog, RAGe, Response Filtering, Dialog Mgmt LLM, BTP subaccount topology, SAP Build Process Automation, SAP Connectivity Service, SAP-delivered vs Custom Skills, 3rd-party IdP, S/4HANA PCE, Capability Patterns, Characteristics)
1.2	March 2026	Sajiv Francis	Added Related Architecture Patterns section (three-pattern model: Joule, ECA/Azure, iGPT); cross-references to Selection Guide

Classification: Internal Use
Review Cycle: Quarterly
Next Review: June 2026

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SAP Joule AI Architecture Overview

End of Document